

Software Test Plan (STP) — Task Manager Application

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Team Details

S. No.	USN	Team Member	Project
9	PES2UG25CS804	Balaji Gowda S R	Task Manager Application
10	PES2UG24CS016	AARYAN DUBEY	Task Manager Application
11	PES2UG24CS006	AADI KORE	Task Manager Application
12	PES2UG24CS041	AKASH PAWAR	Task Manager Application

Revision History

Version	Date	Author	Change Summary
1.0	15-09-2026	Project Team	First issue of the test plan, derived from SRS v1.0 (09-09-2026) and SAD v1.0 (14-09-2026).

1. Introduction

1.1 Objective

This test plan describes how the Task Manager Application (v1.0) will be verified. It states the test goals, coverage boundaries, overall method, needed resources, timelines and accountabilities. The plan is drawn from the team's Software Requirements Specification and the Software Architecture and Design record, so every verification activity can be traced back to an agreed requirement.

1.2 Coverage

Verification addresses the SRS functional scope: sign-in with valid credentials, rejection of invalid credentials, sign-out with session closure; task creation, modification and confirmed removal; priority assignment, due-date handling, workflow status updates and category or label allocation; keyword search, multi-field filtering, deterministic sorting and overdue identification; configured reminders, status-wise summary counts and admin-only operations. Non-functional and security expectations are also verified: common actions within 3 seconds for at least 90% of transactions under normal load, at least 99% availability outside planned maintenance, credential protection, record persistence across restart, usable and accessible UI, authenticated access, role-based admin control, secure password handling, protected transport and input sanitisation. External providers are exercised only at their interface.

1.3 References

- SRS, Task Manager Application, v1.0, 09-09-2026.
- SAD Specification, Task Manager Application, v1.0, 14-09-2026.

- RTM maintained inside the SRS and SAD.

1.4 Terminology

Task: trackable work item. User: signed-in person. Priority: ranking. Status: workflow state. Due date: deadline. Reminder: scheduled notice. UI, CRUD, RBAC as usual.

2. Items Under Test

- Authentication & Session handling.
- Task Service with validation and authorisation.
- Domain rules for priority, due-date, status, category and overdue logic.
- Search & Filter plus monitoring logic.
- Notification Service and Summary Service.
- Administration and access-control paths.
- Persistent Data Store.

3. Functions Selected for Verification

- TMA-F-001, F-002, F-003: valid sign-in, invalid rejection, sign-out.
- TMA-F-010, F-011, F-012: add, amend, confirmed removal.
- TMA-F-020, F-021, F-022, F-023: priority, due date, status, category.
- TMA-F-030, F-031, F-032, F-033: search, filter, sort, overdue flag.
- TMA-F-040, F-041, F-042: reminders, summary tallies, admin gate.
- TMA-NF-001: 90th percentile within 3 s.
- TMA-NF-002: at least 99% reachability, planned downtime excluded.
- TMA-NF-003: credentials shielded, absent from logs.
- TMA-NF-004: saved records survive restart.
- TMA-NF-005: legible, navigable, accessible UI.
- TMA-SR-001 to SR-005: auth gate, RBAC, hash-only passwords, protected transport, input checks.

4. Functions Explicitly Excluded

- Third-party gateway internals — interface only.
- Future calendar/sync internals — boundary only.
- Database/hosting product internals.
- OS/browser faults not caused by the app.

5. Test Method

5.1 Tiers

- Module checks.
- Integration checks (Task+Store, Notification+rules).

- End-to-end system flows.
- Acceptance runs against SRS suites.

5.2 Varieties

- Functional, regression, load, usability/accessibility, and security probes.

Entry rule: stable build delivered, seed data ready, rig prepared.

Exit rule: all planned cases run, no critical/major fault open, SRS benchmarks met.

5.3 Focused security checks

- Unauthenticated task access refused (SR-001).
- Admin screens reject standard users (SR-002, F-042).
- Hash-only password storage (SR-003, NF-003).
- Protected transport (SR-004).
- Hostile input rejected/neutralised (SR-005).
- Sessions end cleanly on sign-out/expiry (F-003).

6. Test Bed

- Hardware: ordinary desktops/laptops/mobiles.
- Software: build under test, supported browser/client, app server, store.
- Tooling: UI automation (e.g. Selenium), API client (e.g. Postman), load tool (e.g. JMeter), tracker (e.g. Jira).
- Data: seeded accounts/tasks/reminders.

7. Timetable

Milestone	Planned date
Test design complete	18-Sep-2026
Environment ready	20-Sep-2026
Execution begins	22-Sep-2026
Execution ends	29-Sep-2026
UAT window	30-Sep-2026 to 02-Oct-2026

8. Outcomes to Hand Over

- This plan, cases (manual+automated), scripts, data, logs, fault reports, summary report.

9. Duties

Role	Name	Duty
QA Lead	AADI KORE	Owns plan, steers verification.
Test Engineer	AARYAN DUBEY	Authors/runs cases, logs faults.

Role	Name	Duty
Test Engineer	AKASH PAWAR	Authors/runs cases, logs faults.
Developer	Balaji Gowda S R	Repairs and retests.
Product Owner	Course Coordinator	Signs off readiness.

10. Risks and Countermeasures

Risk	Countermeasure
Late stable build	Early smoke builds.
Rig outage	Mirrored local fallback.
Flaky reminder gateway	Early interface agreement plus stubs.
Thin seed data	Build and cross-check ahead.

11. Premises and External Needs

- Store steady and reachable.
- Seeds arranged before runs.
- Reminder channel present at least as stub.
- Admin enabled in test deployment.

12. Pause and Restart Rules

Pause when the rig is unreachable beyond 4 hours, or instability blocks over 30% of cases. Restart once blocking faults are fixed and the rig is steady.

13. Trace Handling and Illustrative Cases

Coverage follows the SRS/SAD RTM. Examples: F-001 -> TC-Auth-01; F-010 -> TC-Task-01; F-042+SR-002 -> TC-Admin-01; NF-001 -> TC-Perf-01; SR-003 -> TC-Sec-03.

Case	Req.	Action	Anticipated outcome
TC-Auth-01	TMA-F-001	Valid sign-in.	Dashboard session opens.
TC-Auth-02	TMA-F-002	Wrong password.	Refused with message.
TC-Auth-03	TMA-F-003	Sign out then revisit protected page.	Fresh sign-in demanded.
TC-Task-01	TMA-F-010	Add task with title/notes.	Persists and lists.
TC-Task-02	TMA-F-011	Amend owned task.	Saves correctly.
TC-Task-03	TMA-F-012	Remove after confirm.	Gone from active views.
TC-Org-02	TMA-F-021	Set/revise due date.	Stores and renders.
TC-Search-01	TMA-F-030	Keyword lookup.	Only accessible matches.

Case	Req.	Action	Anticipated outcome
TC-Monitor-01	TMA-F-033	Pass a pending deadline.	Flagged overdue.
TC-Notify-01	TMA-F-040	Trigger eligible reminder.	Notice produced.
TC-Summary-01	TMA-F-041	Open status summary.	Tallies match records.
TC-Admin-01	TMA-F-042	Standard user opens admin.	Refused by role check.
TC-Perf-01	TMA-NF-001	Common screens under load.	90th percentile ≤ 3 s.
TC-Sec-03	TMA-SR-003	Inspect credential fields.	Hashes only.

14. Measures and Communication

Indicators: cases run, pass/fail split, fault density, fault age, coverage. Artefacts: daily note plus closing summary report.

15. Sign-off

Role	Name	Signature / Date
QA Lead		
Dev Lead		
Product Owner		